

AlgoShield AI

AI-Powered Blockchain Security and Monitoring Platform

INTRODUCTION

- AlgoShield AI is an AI-powered blockchain security and monitoring platform designed to improve safety, transparency, and trust in Web3 ecosystems.
- Continuously monitors smart contracts in real time to detect suspicious behavior and vulnerabilities before they cause major problems.
- Provides intelligent AI-based prediction and suggestion models to help developers make better security decisions.
- Offers a developer SDK that integrates easily into existing blockchain applications.

PRODUCT OVERVIEW

- Works as an intelligent security layer combining AI analysis, real-time monitoring, predictive analytics, and smart alert systems.
- Detects abnormal blockchain activities, monitors smart contracts continuously, and predicts security threats.
- Delivers AI-based suggestions and real-time alerts and analytics to developers and users.

WHO ARE THE USERS?

- Blockchain Developers — monitor smart contracts, analyze transaction behavior, receive instant security alerts
- Web3 Startups — improve platform security, reduce risks, and increase user trust
- Smart Contract Projects — continuously monitor contract activity and identify vulnerabilities
- Security Teams — reduce manual monitoring effort using AI-powered threat detection
- Investors and DeFi Users — safer blockchain experiences through scam detection and security insights

BUSINESS IDEA

- Create an AI-driven blockchain security ecosystem protecting projects and users from scams, exploits, and abnormal activities.
- Unlike traditional tools that only display raw blockchain data, AlgoShield uses AI to analyze behavior patterns, predict risks, and provide smart recommendations.
- Platform acts as a blockchain monitoring solution, smart contract security layer, AI-powered risk analysis system, and developer SDK — all in one.

REAL WORLD USEFULNESS

- Prevents scams and suspicious blockchain activities before major financial losses occur
- Provides real-time monitoring and instant alerts for risky behavior
- Uses AI prediction models to identify possible future threats proactively
- Generates intelligent suggestions and recommendations for developers and users
- Reduces manual monitoring work through AI automation
- Improves security, transparency, and trust across Web3 ecosystems

MAIN FEATURES

- AI Security Scanner — Random Forest ML model + 8 TEAL-specific vulnerability rules with severity-weighted Security Score (0–100) and per-line vulnerability mapping
- RAG Fix Suggestions — Phi-3-mini SLM + ChromaDB for line-by-line AI code fix recommendations
- ARC-69 NFT Security Certificates — minted on Algorand testnet for contracts scoring ≥ 70 , serving as verifiable on-chain proof of audit
- 24/7 Live Monitoring & Alerts — Isolation Forest anomaly detection via APScheduler with real-time Email and Telegram Bot notifications

TECHNOLOGIES USED

Layer	Technology
Frontend	React (Vite), TypeScript, Tailwind CSS, Pera Wallet Connect
Backend	FastAPI (Python 3.9+), APScheduler, Motor (async MongoDB)
AI / ML	Scikit-learn (Random Forest, Isolation Forest), Phi-3-mini (llama-cpp), ChromaDB
Blockchain	Algorand Python SDK, ARC-69 NFT Standard, Algonode Indexer API
Databases	MongoDB Atlas (scans, certs) + Supabase / PostgreSQL (monitoring)
SDK / CLI	Node.js, Axios

FUTURE SCOPE

- Multi-chain blockchain support (Ethereum, Solana) with chain-specific ML models
- Advanced AI fraud detection systems with deeper behavioral analysis
- Enterprise-level blockchain monitoring with SLA and white-label options
- Automated incident response systems triggered by real-time anomaly detection
- AI-powered threat intelligence and advanced analytics and reporting dashboards

HOW AlgoShield AI IS DIFFERENT FROM Immunefi?

- Built Specifically for Algorand - AlgoShield AI is focused on Algorand smart contract security, while Immunefi is mainly a multi-chain bug bounty platform.
- Instant AI-Based Security Scoring - AlgoShield AI automatically generates vulnerability reports and security scores instead of relying only on manual bug submissions.
- On-Chain NFT Security Proof- AlgoShield AI provides NFT-based audit certificates that publicly verify a contract's security assessment.
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